

Milo Shan

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github.com/MoodyMarshmallow

Education

University of Pennsylvania, BS in Artificial Intelligence Aug 2024 – May 2028

- Relevant Courses: Applied Machine Learning, Deep Learning Lab, Human Computer Interaction

Experience

Software Engineering Intern, Pattern – Lehi, UT May 2026 - Present

- Embedded as an Amazon copywriter and content manager to map end-to-end content workflow; presented findings to Data Science leadership and catalyzed team-wide pivot from maintaining legacy tools to building shared AI-agent infrastructure.
- Co-developed 9 platform-agnostic tools exposing existing copy-generation pipeline to Claude Cowork and an internal chatbot, cutting end-to-end turnaround from 4–8 weeks to 3–4 days.
- Leading alpha-testing with ~20 users, translating feedback into product requirements for PMs and shipping 10+ fixes across 3 repositories.

Lab Member, ChatGPT Lab, OpenAI – New York, NY May 2025 - Present

- Contributed to ChatGPT Study Mode development, serving 700M+ weekly users through feature testing.
- Invited to interview with CRO in technical panels on product strategy.

Researcher & Backend Lead, PennNLP, UPenn – Philadelphia, PA May 2025 - Aug 2025

- Led 7-person team in developing multi-agent interaction platform, architecting and implementing 5,000+ line Python backend.
- Engineered game-like, turn-based simulation environment using FastAPI and in-house framework for multi-agent interaction research
- Built an agent performance evaluation system using PyTest
- Developed visualization layer using Godot Engine for interactive simulation monitoring and debugging PyTest.

Publications

Computational prediction of MOFs with the potential to improve the efficiency of industrial CO₂ capture July 2024

Machine Learning Regression to Predict Soil Moisture in Domestic Garden Environments May 2023

Projects

NanoGPT Training Speedrun [Github link](#)

- Optimized training of nanoGPT on ~900M tokens of FineWeb dataset for speed and accuracy.
- Implemented and tested gated attention variants to an existing speedrun record.
- Achieved 0.6% accuracy improvement.

Fintuning of DINOv3 for Image2GPS Task [Github link](#)

- Designed preprocessing pipeline cleaning ~3,500 self-collected phone-photo metadata rows; delegating them to ~90 contiguous regions via rasterization, k-medoids, and custom growth algorithm.
- Fine-tuned DINOv3-S classifier using soft adjacent-region labels & top-5 weighted-centroid decoding for final lat-long prediction. Achieved 37.27 m external-test error.

Technical Skills

Languages: Python, C++, Java, C#, TypeScript, CSS, SQL

Frameworks & Technologies: PyTorch, React, Next.js, FastAPI, Tailwind CSS, Godot Engine

Tools & Platforms: PostgreSQL, Vite, Supabase, Linux, Jupyter, Arduino, Git